TRUTHLENS AI VERIFICATION REPORT

AI-Powered Misinformation Detection & Forensic Claim Analysis System

Article Title: NASA James Webb Space Telescope Observes Exoplanet Atmosphere	**Final Verdict:** **INCONCLUSIVE / MIXED EVIDENCE**
Source URL / Domain: Direct Submission	**Credibility Index: 60.6 / 100**
Analyzed On: 04 November 2026, 08:39 PM	**Primary Category:** Science

1. Multi-Model Artificial Intelligence Detection

Model Architecture	Classification Verdict	Genuine Confidence	Fake Confidence
Logistic Regression (TF-IDF)	Genuine	66.4%	33.6%
Multinomial Naive Bayes	Genuine	99.7%	0.3%
Random Forest Classifier	Genuine	68.7%	31.3%
DistilBERT Contextual Transformer	Inconclusive	46.6%	53.4%

2. Explainable AI (XAI) Influential Terms Attribution

Influential Word / Token	Attribution Weight	Directional Impact
carbon	0.304	**Genuine**
space	0.288	**Genuine**
models	0.162	**Genuine**
james	0.143	**Genuine**
webb	0.143	**Genuine**
telescope	0.143	**Genuine**

3. Source Credibility & Fact-Check Verification

Domain Reputation: Unspecified Direct Text — Unverified Source (Score: 50.0/100)
Details: Submitted as direct raw text without a verifiable online publisher domain.

Corroborating Evidence Found:
• **NASA Goddard Space Flight Center & Nature:** Spectroscopic data from the JWST NIRSpec instrument definitively confirmed atmospheric molecular signatures including carbon dioxide and water vapor. *(VERIFIED / ACCURATE)*

4. Emotional Tone & Sensationalism Analysis

Sentiment: Neutral / Objective (Polarity: 0.0) | **Dominant Emotion:** Surprise | **Clickbait Index:** 8/100 (Standard Journalistic Phrasing)

Methodological Notice & Ethical AI Disclaimer: *TruthLens AI employs statistical NLP models, deep learning contextual representations, and external fact databases to estimate credibility. Results reflect probabilistic machine evaluation and available external evidence. This report should be utilized as an assistive decision-support tool rather than an infallible declaration of absolute truth.*